IMX6ULLEVKHUG

i.MX 6ULL EVK Board Hardware User Guide

Rev. 2 — 2 November 2023 User guide

Document information

Information	Content
Keywords	IMX6ULLEVKHUG, i.MX 6ULL, application processors, board user guide
Abstract	This document is the hardware user guide for the i.MX 6ULL evaluation kit (EVK) based on the NXP i.MX 6ULL applications processor.

1 Introduction

This document is the hardware user guide for the i.MX 6ULL evaluation kit (EVK) based on the NXP i.MX 6ULL applications processor. This user guide includes system setup and debugging. It provides detailed information on the overall design and usage of the EVK board from a hardware systems perspective.

1.1 Board overview

The EVK board is a platform designed to showcase commonly used features of the i.MX 6ULL applications processor in a small, low-cost package. The i.MX 6ULL EVK board is an entry level development board. It gives the developer an option of becoming familiar with the processor before investing a large amount of resources in more specific designs.

The features of the i.MX 6ULL EVK board are shown in Table 1.

Table 1. Board features

Processor:	NXP Applications Processor	MCIMX6Y2DVM09AB
DRAM Memory:	Micron 4 Gb DDR3L SDRAM	MT41K256M16TW-107:P
DCDC:	MPS	MP2144GJ and AP3401KTTR-G
LDO	RICHTECK	RT9169-33GVL RT9169-18GVL RT9193-33GB
	UNION	UM1750S-00
Mass storage	MicroSD card connector	
	256 Mbit QSPI Flash	
	32 GB NAND Flash (unpopulated)	
	32 GB eMMC (unpopulated)	
Display interface	LCD connector	
	HDMI connector (unpopulated)	
Ethernet	Dual RJ45 Connector for 10/100 Base-T	
USB	One USB 2.0 Standard-A host connector	
	One USB 2.0 Micro-B OTG connector	
Audio connectors:	3.5 mm Stereo Headphone output	
	Mono-Microphone input on board	
	Left and Right Speaker Out connectors	
Power connector:	5 V DC-Jack, 2.0 mm (Power supply: DC5V/4A, 2.1 mm X 5.5 mm)	
Debug connectors:	UART to USB connector	
	RJ-45 connector for 10/100 Base-T	
	20-pin Standard JTAG connector	
Sensor	eCOMPASS (unpopulated)	
	Accelerometer	
	Gyroscope (unpopulated)	
Camera	CMOS Camera connector(unpopulated)	
CAN	CAN bus connector	

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Table 1. Board features*...continued*

Bluetooth	20-pin Bluetooth Connector
User Interface Buttons	ON/OFF, Reset button
LED Indicators	Power status, UART
Coin Cell	Connection point for ML414 Coin Cell (unpopulated) for RTC Operation
Coin Cell	Connection point for ML414 Coin Cell (unpopulated) for RTC Operation
Expansion Port	ARDUINO Headers
PCB	Main board: 2.66 inch x 1.27 inch (6.76 cm x 4.24 cm), 4-layer board
	Base board: 5.12 inch x 4.25 inch (13.0 cm x 10.8 cm), 4-layer board

1.2 i.MX 6ULL EVK contents

The i.MX 6ULL EVK contains the following items:

- i.MX 6ULL main board and base board
- Quick start guide

1.3 i.MX 6ULL EVK board revision history

The board assembly version is printed on a label, attached to the bottom side. The assembly version is the letter designation following the schematic revision: base board is 700-28616 REVE, main board is 700-29364 REVB.

2 Specifications

This chapter provides detailed information about the electrical design and practical considerations of the EVK board. The chapter is organized to discuss each block in the following block diagram of the EVK board.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

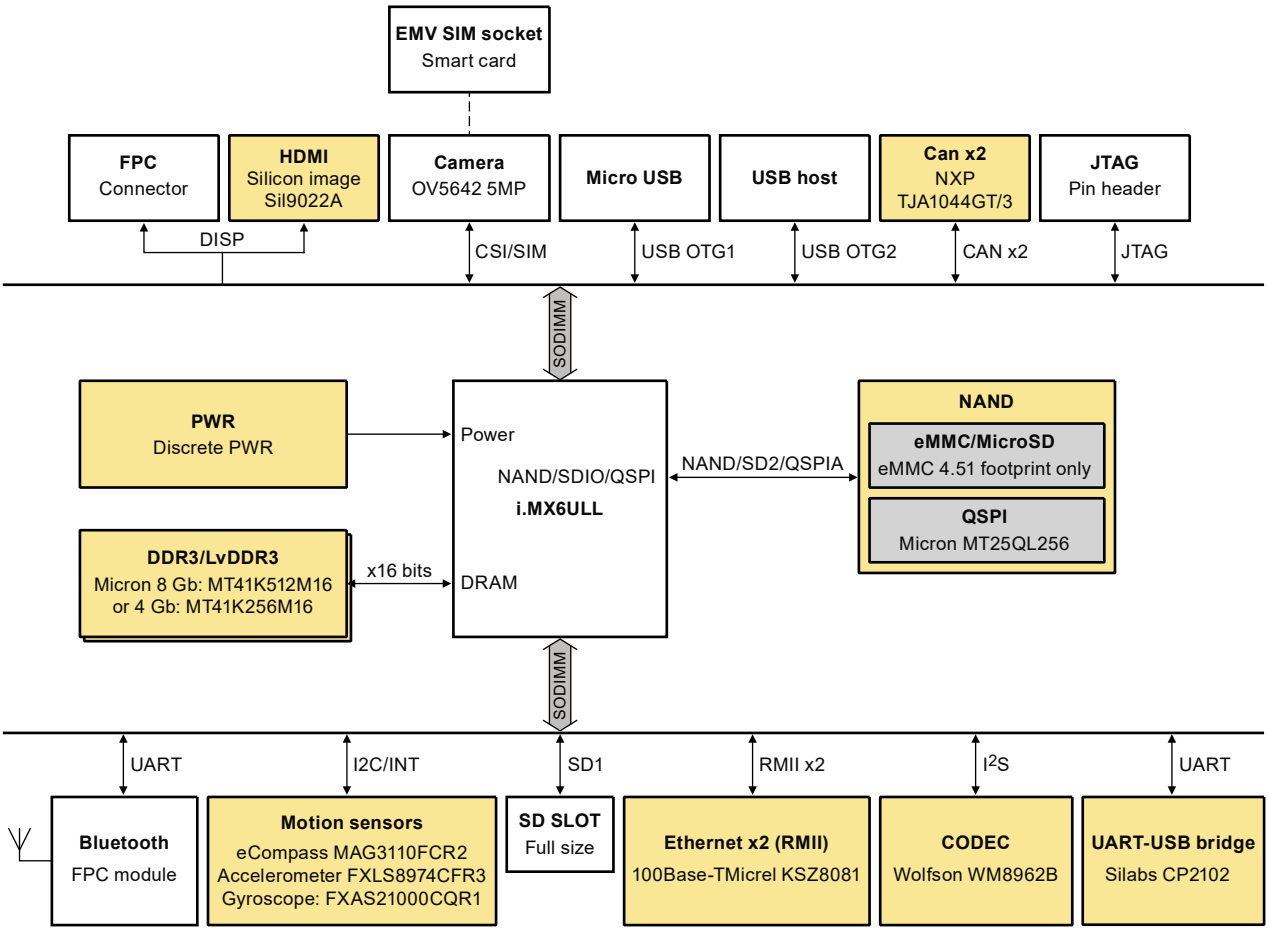


Figure 1. Block diagram

The overview of the i.MX 6ULL EVK board is shown in Figure 2.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

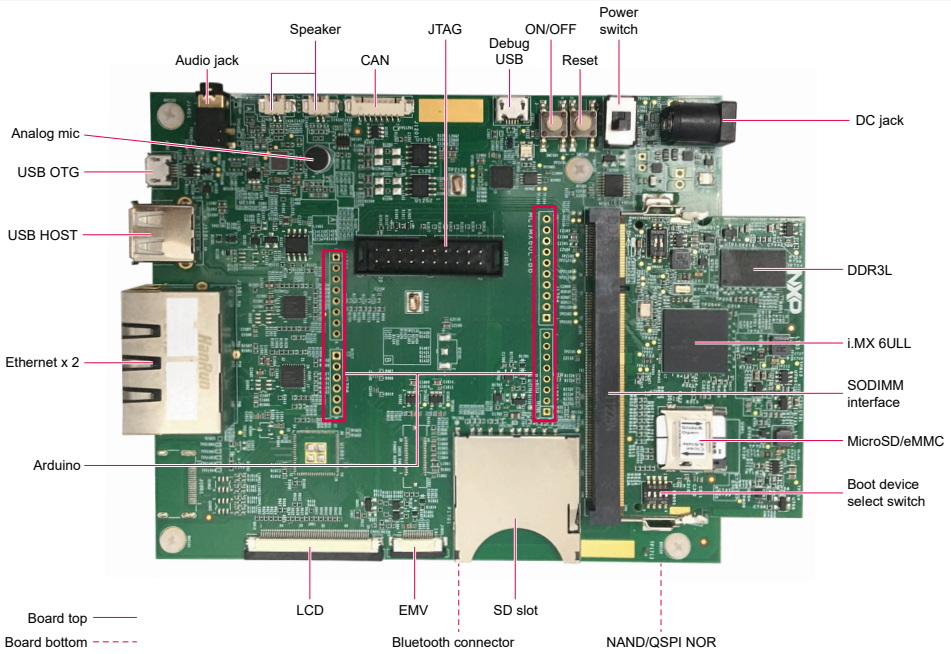


Figure 2. Overview of the i.MX 6ULL EVK board

2.1 i.MX 6ULL processor

The i.MX 6ULL processor is NXP's latest multimedia-focused achievement offering high-performance processing and functional integration. It is aimed at the growing market of connected devices. The i.MX 6ULL processor features NXP's advanced implementation of the single ARM Cortex-A7 core, which operates at speeds up to 900 MHz. The i.MX 6ULL includes an integrated power management module that reduces the complexity of external power supply and simplifies the power sequencing. Each processor provides a 16-bit DDR3/LVDDR3/LPDDR2 memory interface and several other interfaces for connecting peripherals, such as WLAN, Bluetooth, GPS, and displays.

More detailed information about the processor can be found in the data sheets and reference manual listed in Section 6.

2.2 Boot mode configurations

BOOT_MODE [1:0] is used to select system boot mode. On the i.MX 6ULL EVK board, a dual-switch (SW602) is used to select the input voltage of these two pins, either 0 V or 3.3 V.

Table 2. SW602 boot mode pin settings

D1/MODE1	D2/MODE0	Boot Mode
OFF	OFF	Boot from Fuses
OFF	ON	Serial Downloader
ON	**OFF**	**Internal Boot**
ON	ON	Reserved

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.



Figure 3. SW602 boot mode settings

Typically, the internal boot is selected for normal boot, which is configured by external BOOT_CFG GPIOs. Then the developer must set the switch SW602: D2(MODE0) to OFF, D1(MODE1) to ON, and use SW601 to select the boot device. This is shown in Table 3 and Figure 4.

Table 3. SW601 boot device settings

D1	D2	D3	D4	Boot Device
OFF	**OFF**	**ON**	**OFF**	**MicroSD**
OFF	OFF	OFF	OFF	QSPI
OFF	ON	ON	OFF	EMMC
ON	ON	OFF	ON	NAND

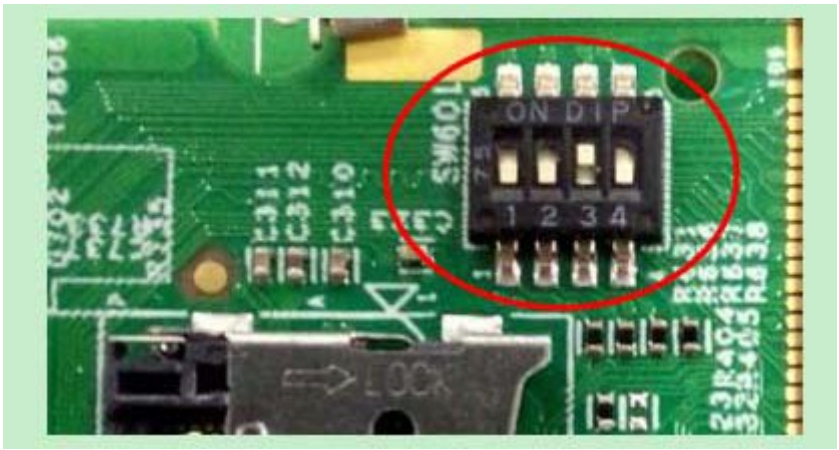


Figure 4. SW601 boot device settings

On the i.MX 6ULL EVK board, the default boot device is a USDHC interface. There are two SD slots on the board, one MicroSD (USDHC-2) on the main board and one standard SD (USDHC-1) on the base board. MicroSD (USDHC-2) is the default boot device. To boot from the SD (USDHC-1) by default, R613 (10K) must be removed and R642 (10K) must be installed. The location of the R613 and R642 is on the bottom side, as shown in Figure 5.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

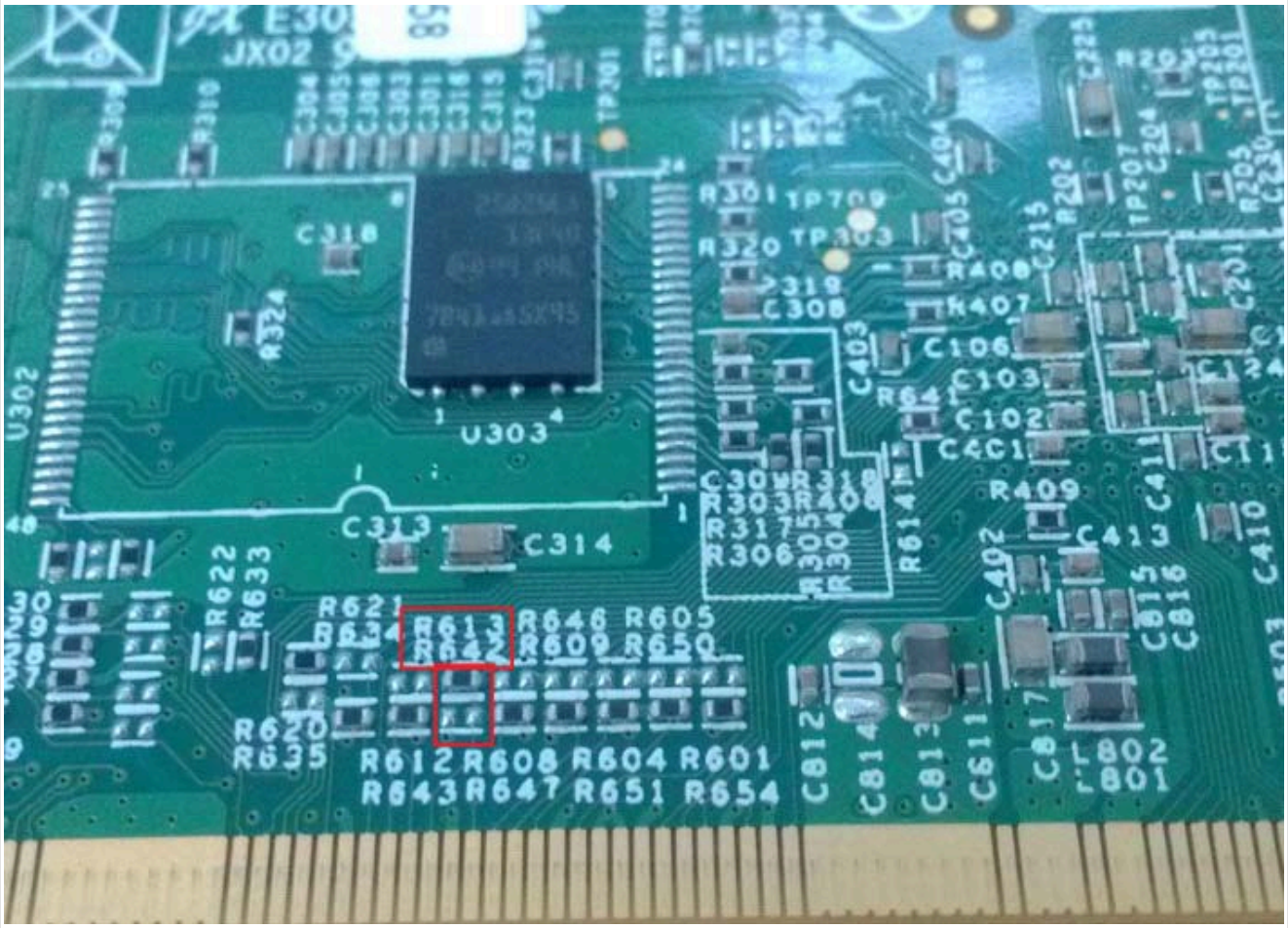


Figure 5. Location of the R613 and R642

Note:

For more information about boot mode configuration, see the System Boot chapter of the i.MX 6ULL Reference Manual.

For more information about i.MX 6ULL EVK boot device selection and configuration, see the main board schematic.

2.3 Power tree

A DC 5 V/4 A external power supply with a 2.1 mm x 5.5 mm plug is used to supply the i.MX 6ULL EVK board at connector J2001, and a slide switch SW2001 is used to turn the power ON/OFF. There is an over-voltage protection circuit at the input power rail. So, if the output voltage of the power supply exceeds 5.6 V, the circuit shuts down the system. Discrete regulators are used to generate different power rails for the whole system.

The power tree is shown in Figure 6.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

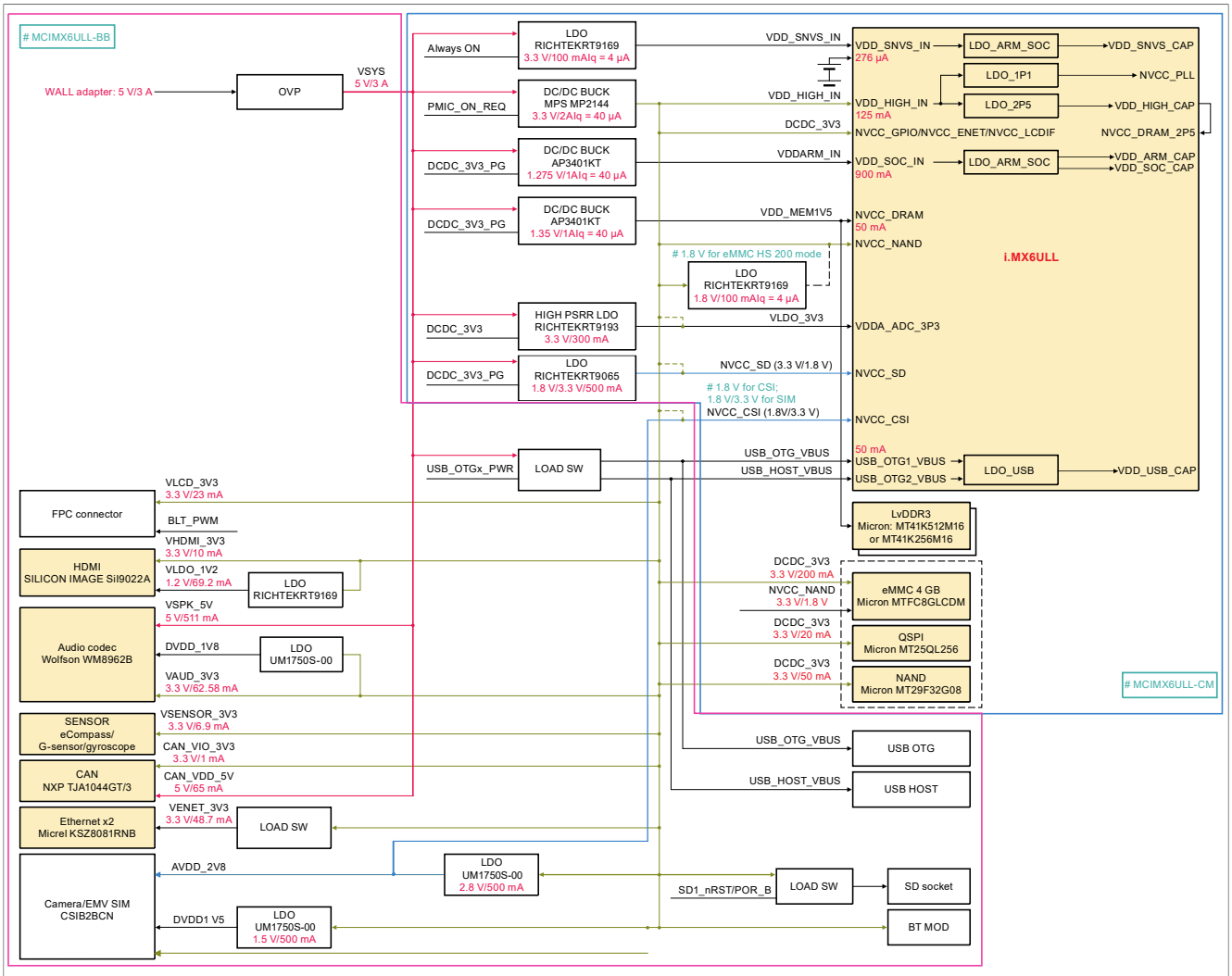


Figure 6. Power tree

The power tree displays all the power supply rails used on the EVK board.

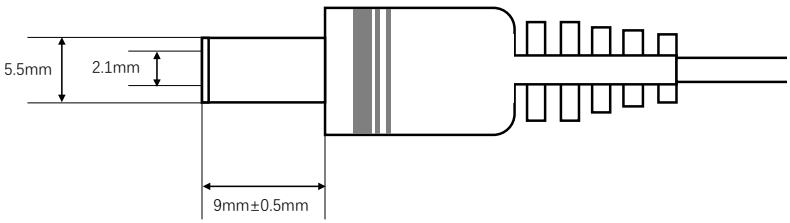


Figure 7. Plug size for DC5V power supply

The power control logic of the i.MX 6ULL EVK board is shown in Figure 8.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

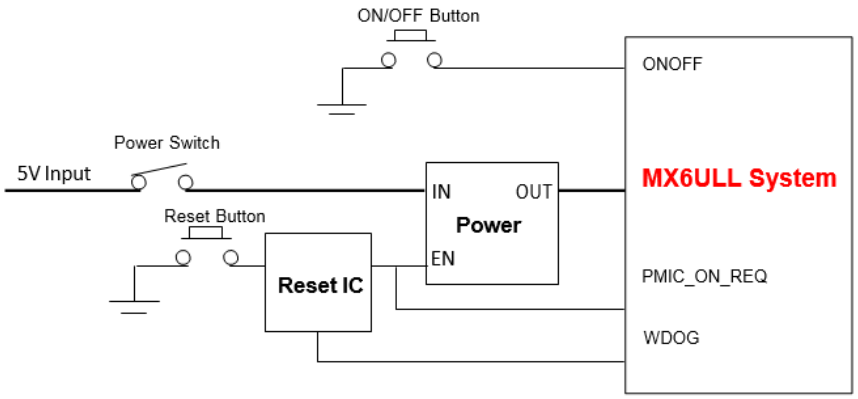


Figure 8. Power control diagram

The power rails on the board are shown in Table 4.

Table 4. Power rails

Regulator	Voltage/max-current	Power domains	Modules
DC/DC BUCK AP3401KTTR-G	1.3 V / 1 A	VDD_ARM_SOC_IN	VDDSOC
DC/DC BUCK AP3401KTTR-G	1.35 V / 1 A	NVCC_DRAM	NVCC_DRAM DDR3L SDRAM
DC/DC BUCK MP2144	3.3 V / 2 A	VDD_HIGH_IN DCDC_3V3	VDD_HIGH GPIO ENET module LCD module NAND module QSPI module CAN module Audio module HDMI module Camera module Bluetooth module
LDO RT9169	3.3 V / 100 mA	VDD_SNVS_IN	VDD_SNVS_IN
LDO RT9193	3.3 V / 300 mA	VLDO_3V3	ADC module
LDO UM1750	1.8 V or 3.3 V / 300 mA	NVCC_SD	SD module
LDO UM1750	2.8 V / 300 mA	NVCC_CSI	CSI module
LDO UM1750	1.5 V / 300 mA	DVDD_1V5	Camera module
LDO RT9169	1.2 V / 100 mA	VLDO_1V2	HDMI module

2.4 DDR3L DRAM memory

The i.MX 6ULL EVK board has one 256 Meg x 16 DDR3L SDRAM (MT41K256M16TW-107:P) for a total of 512 MB RAM memory.

The DDR_VREF is created by a simple voltage divider using 1.5 K ohm 1 % resistors and 0.1 µF capacitors for stability. The relatively small value resistors provide enough current to maintain a steady midpoint voltage. The calibration resistors used by the DDR3L chips and the processor are 240 ohm 1 % resistors. These resistor values are specified by the DDR3L specifications.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

2.5 SD card slots (J301, P1701)

There are two SD card connectors (J301, P1701) on the i.MX 6ULL EVK board, which are as follows:

- J301 on the i.MX 6ULL EVK Main board is the Micro SD slot for a USDHC2 interface.
- P1701 on the i.MX 6ULL EVK base board is a standard SD slot for USDHC1. It can also support accessories such as a WiFi card.

2.6 eMMC

The eMMC interface is connected to the USDHC2 of i.MX 6ULL. It can support up to eMMC 4.5, the eMMC device is not populated by default on the EVK board. To boot from eMMC, remove the Micro SD socket and populate the eMMC device. Then, change the boot device switch (SW601) settings (OFF, ON, ON, OFF) as shown in Table 3.

Also, the BOOT_CFG settings must be adjusted for the specified eMMC device by exchanging the pull up/down resistors. See the main board schematic for more information.

2.7 QSPI Nor flash

On the 6ULL EVK board, there is one 256 MB QSPI Nor Flash device. To boot from the QSPI flash, the boot device switch (SW601) settings must be changed to OFF, OFF, OFF, OFF, as shown in Table 3.

Also, the developer must adjust the BOOT_CFG settings for the specified QSPI Nor Flash device, by exchanging the pull up/down resistors. See the main board schematic for more information.

2.8 NAND flash

NAND flash is not populated by default on the i.MX 6ULL EVK board. To boot from the NAND flash, the following reworks must be implemented:

1. Remove the QSPI Nor flash device
2. Populate the NAND device
3. Change the Boot-mode switch (SW601) settings to ON, ON, OFF, ON
4. The developer must adjust the BOOT_CFG settings for the specified NAND Flash device, by exchanging the pull up/down resistors. See the main board schematic for more information.

Note: *The data lines of NAND flash are multiplexed with the eMMC and MicroSD, so they cannot be used at the same time.*

2.9 Dual Ethernet connector (J1501)

There are two Ethernet MAC controllers in the i.MX 6ULL processor. The Ethernet subsystem of the i.MX 6ULL EVK board is provided by the MICREL KSZ8081RNB 10/100M Ethernet Transceiver (U1501, U1601) and a dual port RJ45 (J1501) with integrated Magnetic. The MAC addresses are shown on the sticks on the RJ45 connector.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

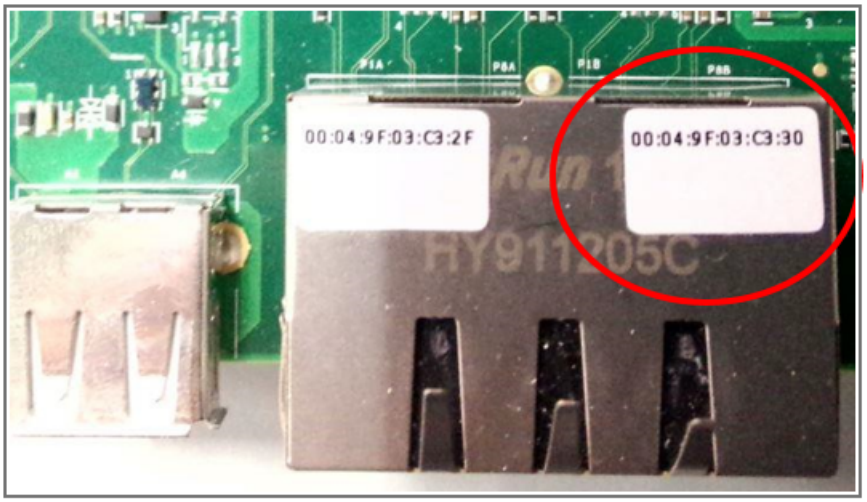


Figure 9. Dual Port RJ45

2.10 USB PHY connector (J1102, J1101)

The i.MX 6ULL Applications Processors contain two high speed (HS) USB 2.0 OTG (Up to 480 Mbps) controllers, with two integrated HS USB Phy. On the EVK board, J1101 is a Type-A connector for USB host port and J1102 is a Micro-B connector for USB OTG port. The OTG port J1102 is also used for downloading the boot image by MFGTool in Serial download Mode.

2.11 Audio input/output connectors (J1401/P1401/J1402 and J1403)

The Audio CODEC used on the i.MX 6ULL EVK board is Wolfson's low-power, high-quality Stereo Codec, WM8962B. The digital interface between i.MX 6ULL and WM8962B includes four signals:

- SYNC_CLK
- BCLK
- DACDAT
- ADCDAT

The i.MX 6ULL also provides the MCLK to WM8962B.

i.MX 6ULL EVK includes one headphone interface (J1401), one onboard MIC (P1401), and two speaker interfaces (J1402, J1403). J1401 is a 3.5 mm 4-pole (or TRRS) phone jack, which supports jack detect.

2.12 USB debug connector (J1901)

On the EVK board, CP2102, a USB to Serial UART IC is used to convert the UART signal to the USB signal. With the Micro-B USB connector J1901, the connection for debugging is simplified. A Micro-B to standard A USB cable can be used.

UART1 port is used as the default debug port. No RTS or CTS signals are sent from the processor to the debug connector because these signals are commonly ignored by most applications. The required terminal settings are shown in Table 5.

Table 5. Terminal settings

Baud rate	115,200
Data bits	8

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Table 5. Terminal settings...*continued*

Parity	None
Stop bits	1
Flow control	None

2.13 Bluetooth Connector (J1701)

On the EVK board, the J1701 is connected to i.MX 6ULL UART2 port, and can be used for Bluetooth. Silex Bluetooth module SX-SDCAN-2830BT is recommended. To verify other Bluetooth modules, a convert board may be needed.

Note: *J1701 is a bottom-contact type connector for BT connection. A 20-pin, 0.5mm pitch FPC cable is needed for connecting Silex Bluetooth module SX-SDCAN-2830BT and EVK, as shown in Figure 10.*



Figure 10. Connect the BT module

2.14 JTAG connector (J1902)

J1902 is a standard 20-pin/2.54 mm Box Header Connector for JTAG. The pin definitions are shown in Figure 11.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

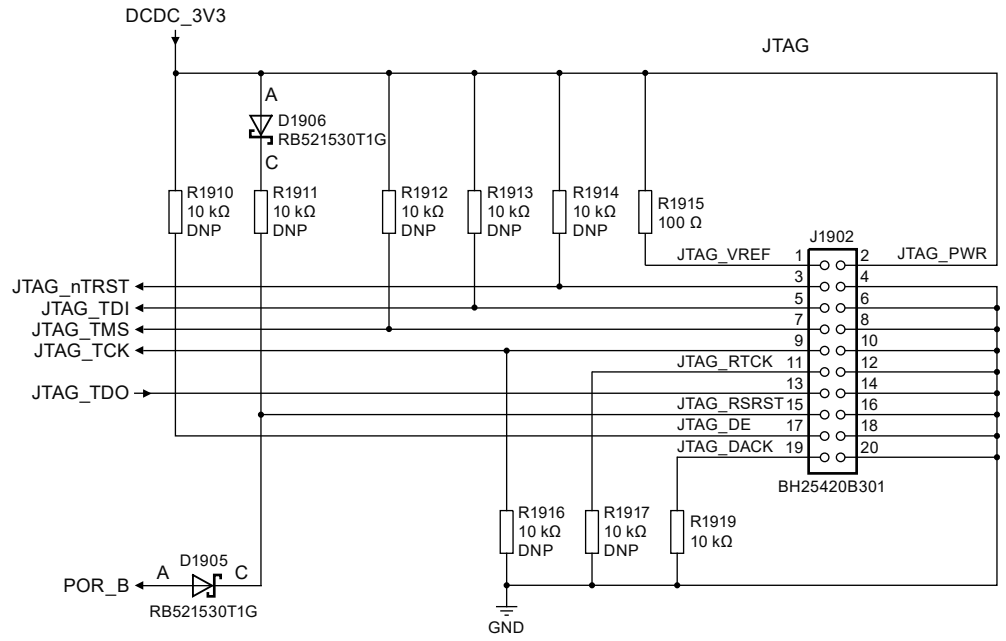


Figure 11. JTAG pin definitions

Note: *JTAG is dedicated with Audio CODEC signals. Remove R1407, R1431~R1434 resistors while using JTAG debug.*

2.15 ARDUINO expansion port (J1703-J1706)

J1703-J1706 (unpopulated) is defined as the ARDUINO Interface. To use the processor for ARDUINO development, the following rework must be implemented.

- Populate- R1717-R1732

The pin definitions of the ARDUINO interface are shown in Table 6 and Table 7.

Table 6. ARDUINO pin definitions for J1703 and J1706

J1703	J1706
UART_RX/D0	A0
UART_TX/D1	A1
D2	A2
D3	A3
D4	A4
D5	A5
D6	
D7	

Table 7. ARDUINO pin definitions for J1704 and J1705

J1704	J1705
D8	NC

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Table 7. ARDUINO pin definitions for J1704 and J1705*...continued*

J1704	J1705
D9	3.3 V
D10/SPI_CS	RESET
D11/SPI_MOSI	3.3 V
D12/SPI_MISO	5 V
D13/SPI_CLK	GND
GND	GND
3.3 V	3.3 V
D14/I2C_SDA	
D15/I2C_SCL	

2.16 User interface switches

There are three user interface switches on the EVK board. Their functionality is as follows.

2.16.1 ON/OFF button (SW2101)

SW2101 is the ON/OFF button for the i.MX 6ULL EVK board. The function of this button is listed below.

- In the Yocto project, short press and long press only generate an interrupt. The usage is defined by an upper software.
- Long press (>5 sec) can force the system to shut down immediately.
- If the board is in SHUTDOWN state, a short press of the button restarts (boot) the system.
- If the board is in STANDBY state, a short press of the button brings the system out of standby (resume operations, no boot).

2.16.2 Reset

There are two ways to reset the i.MX6ULL EVK board:

1. Hardware way to reset the system using SW2102:
   - Pressing the RESET button (SW2102) in the Power On state forces to reset almost all the power rails except the VDD_SNVS on the i.MX6ULL EVK board. The i.MX6ULL applications processor is immediately turned off, and reinitiates a boot cycle from the Processor Power Off state.
2. Software way to reset the system using WDOG signal:
   - When the system assertsan nWDOG signal, U703 gets shut down, and the system powers down.Then, nWDOG is released, U703 is powered on by PMIC_ON_REQ, and the system then reboots. It is important that the system reset must be power down and then power on, which avoids boot failure caused by DVFS or DRAM (LPDDR2 if used).

2.16.3 Power switch (SW2001)

SW2001 is a slide switch to control the power of the i.MX 6ULL EVK board. The function of this switch is listed below.

- Sliding the switch to the ON position connects the 5 V power supply to the evaluation board main power system.
- Sliding the switch to OFF position immediately removes all power from the board.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

2.17 Sensors

The i.MX 6ULL EVK board has three sensors: Accelerometer (FXLS8974CFR3), eCOMPASS (MAG3110), and Gyroscope (FXAS21002CQ, unpopulated). These three sensors are connected to the i.MX 6ULL I2C1 port.

2.17.1 Accelerometer

FXL58974CFR3 is a small, low-power, 3-axis, linear accelerometer in a 2 mm x 2 mm x 1 mm DFN package. FXL58974CFR3 has dynamically selectable acceleration full-scale ranges of ±2g/±4g/±8g/±16g and 12 bits of resolution.

2.17.2 eCOMPASS (U1302)

MAG3110 is a small, low-power, 3-axis digital magnetometer. It works by measuring the strength of a magnetic field, which is a combination of the earth's magnetic field and the magnetic fields of nearby objects, including distortions. The device can be used with a 3-axis accelerometer to produce orientation-independent, accurate compass heading information.

2.17.3 Gyroscope (U1303)

FXAS21002CQ is a small, low-power, yaw, pitch, and roll angular rate gyroscope with 16-bit ADC resolution. The full-scale range is adjustable from 250/s to 2000/s. It is not populated by default. To verify this feature, the chip must first be purchased and soldered.

2.18 User interface LED indicators

There are two LED status indicators located on the board. The functions of these LEDs include:

- Main power supply (D2003)
  - Green: DC 5 V main supply (WALL_5V_DC_JACK) is normal.
  - Red: WALL_5V_DC_JACK is over 5.6 V.
  - Off: The board is not powered.
- UART (D1903)
  - Green flashing: Debug UART data are being transmitted to the PC.
  - Orange flashing: Debug UART data are being received from the PC.

2.19 Optional back-up coin cell holder (BT2101)

On the i.MX 6ULL EVK board, there is a connector (BT2101) for holding an ML414 Lithium coin cell battery. The coin cell provides an alternative power supply for the i.MX 6ULL VDD_SNVS_IN power rail when the main VDD_SNVS_3V3 is off. When VDD_SNVS_3V3 is on, the coin cell can be charged through a current limit resistor R101. The i.MX 6ULL has an internal LDO to regulate the VDD_SNVS_IN power to supply the RTC subsystem. When the DC 5 V power supply is removed, the coin cell provides power only to the VDD_SNVS_IN power rail of the EVK board. To increase the keep time of RTC, the developer must optimize the power consumption of the whole VDD_SNVS_IN power rail.

3 PCB information

Both the base board and main board are made using standard 4-layer technology. The material used was FR4.

The PCB stack-up information is shown in Table 8.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Table 8. Board stack-up information

Base board				Main board			
Layer	**Description**	**Copper (Oz)**	**Dielectric Thickness (mil)**	**Layer**	**Description**	**Copper (Oz)**	**Dielectric Thickness (mil)**
1	Signal	1	—	1	Signal	1	—
	Dielectric	**—**	**3.0**		**Dielectric**	**—**	**3**
2	GND	1	—	2	GND	1	—
	Dielectric	**—**	**36**		**Dielectric**	**—**	**28**
3	Power	1	—	3	Power	1	—
	Dielectric	**—**	**3.0**		**Dielectric**	**—**	**3**
4	Signal	1	—	4	Signal	1	—

4 EVK design files

The schematics, layout files, and Gerber files (including Silkscreen) can be downloaded from www.nxp.com/iMX6ULLEVK.

5 Contents of the Evaluation Kit

Table 9. EVK contents

Item	Description
Main Board	CPU board with i.MX 6ULL applications processor, memory, discrete power, and microSD card slot
Base Board	Peripherals and connectivity board
Documentation	Quick Start Guide

Table 10. Required preparation equipment

Item	Description
Power Supply	Output: 5 V/4 A; Plug: 2.1 mm x 5.5 mm
USB Cable	USB cable (micro-B to standard-A)
MicroSD card	Bootable Linux image

6 References

The following references can be found on www.nxp.com:

- *i.MX 6ULL Applications Processor Reference Manual* (document IMX6ULLRM)
- *i.MX 6ULL Applications Processors for Consumer Products* (document IMX6ULLCEC)
- *i.MX 6ULL Applications Processors for Industrial Products* (document IMX6ULLIEC)

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

7 Revision history

Section 7 summarizes the revisions to this document.

Revision history

Revision number	Release date	Description
2	2 November 2023	• Updated the following sections: – Section 1.3, Section 2.1, Section 2.11, Section 2.17, Section 2.17.1, LCD interface, Section 5, and Section 6 • Updated the following tables: – Table 1, Table 4, and Table 10 • Updated the following images: – Figure 1, Figure 2, Figure 6, and Figure 7 • Made some editorial changes
1	05/2017	Section 2: changed “500 MHz” to “900 MHz”
0	10/2016	Initial release

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Legal information

Definitions

Draft — A draft status on a document indicates that the content is still under internal review and subject to formal approval, which may result in modifications or additions. NXP Semiconductors does not give any representations or warranties as to the accuracy or completeness of information included in a draft version of a document and shall have no liability for the consequences of use of such information.

Disclaimers

Limited warranty and liability — Information in this document is believed to be accurate and reliable. However, NXP Semiconductors does not give any representations or warranties, expressed or implied, as to the accuracy or completeness of such information and shall have no liability for the consequences of use of such information. NXP Semiconductors takes no responsibility for the content in this document if provided by an information source outside of NXP Semiconductors.

In no event shall NXP Semiconductors be liable for any indirect, incidental, punitive, special or consequential damages (including - without limitation - lost profits, lost savings, business interruption, costs related to the removal or replacement of any products or rework charges) whether or not such damages are based on tort (including negligence), warranty, breach of contract or any other legal theory.

Notwithstanding any damages that customer might incur for any reason whatsoever, NXP Semiconductors' aggregate and cumulative liability towards customer for the products described herein shall be limited in accordance with the Terms and conditions of commercial sale of NXP Semiconductors.

Right to make changes — NXP Semiconductors reserves the right to make changes to information published in this document, including without limitation specifications and product descriptions, at any time and without notice. This document supersedes and replaces all information supplied prior to the publication hereof.

Suitability for use — NXP Semiconductors products are not designed, authorized or warranted to be suitable for use in life support, life-critical or safety-critical systems or equipment, nor in applications where failure or malfunction of an NXP Semiconductors product can reasonably be expected to result in personal injury, death or severe property or environmental damage. NXP Semiconductors and its suppliers accept no liability for inclusion and/or use of NXP Semiconductors products in such equipment or applications and therefore such inclusion and/or use is at the customer's own risk.

Applications — Applications that are described herein for any of these products are for illustrative purposes only. NXP Semiconductors makes no representation or warranty that such applications will be suitable for the specified use without further testing or modification.

Customers are responsible for the design and operation of their applications and products using NXP Semiconductors products, and NXP Semiconductors accepts no liability for any assistance with applications or customer product design. It is customer's sole responsibility to determine whether the NXP Semiconductors product is suitable and fit for the customer's applications and products planned, as well as for the planned application and use of customer's third party customer(s). Customers should provide appropriate design and operating safeguards to minimize the risks associated with their applications and products.

NXP Semiconductors does not accept any liability related to any default, damage, costs or problem which is based on any weakness or default in the customer's applications or products, or the application or use by customer's third party customer(s). Customer is responsible for doing all necessary testing for the customer's applications and products using NXP Semiconductors products in order to avoid a default of the applications and the products or of the application or use by customer's third party customer(s). NXP does not accept any liability in this respect.

Terms and conditions of commercial sale — NXP Semiconductors products are sold subject to the general terms and conditions of commercial sale, as published at https://www.nxp.com/profile/terms, unless otherwise agreed in a valid written individual agreement. In case an individual agreement is concluded only the terms and conditions of the respective agreement shall apply. NXP Semiconductors hereby expressly objects to applying the customer's general terms and conditions with regard to the purchase of NXP Semiconductors products by customer.

Export control — This document as well as the item(s) described herein may be subject to export control regulations. Export might require a prior authorization from competent authorities.

Suitability for use in non-automotive qualified products — Unless this document expressly states that this specific NXP Semiconductors product is automotive qualified, the product is not suitable for automotive use. It is neither qualified nor tested in accordance with automotive testing or application requirements. NXP Semiconductors accepts no liability for inclusion and/or use of non-automotive qualified products in automotive equipment or applications.

In the event that customer uses the product for design-in and use in automotive applications to automotive specifications and standards, customer (a) shall use the product without NXP Semiconductors' warranty of the product for such automotive applications, use and specifications, and (b) whenever customer uses the product for automotive applications beyond NXP Semiconductors' specifications such use shall be solely at customer's own risk, and (c) customer fully indemnifies NXP Semiconductors for any liability, damages or failed product claims resulting from customer design and use of the product for automotive applications beyond NXP Semiconductors' standard warranty and NXP Semiconductors' product specifications.

Translations — A non-English (translated) version of a document, including the legal information in that document, is for reference only. The English version shall prevail in case of any discrepancy between the translated and English versions.

Security — Customer understands that all NXP products may be subject to unidentified vulnerabilities or may support established security standards or specifications with known limitations. Customer is responsible for the design and operation of its applications and products throughout their lifecycles to reduce the effect of these vulnerabilities on customer's applications and products. Customer's responsibility also extends to other open and/or proprietary technologies supported by NXP products for use in customer's applications. NXP accepts no liability for any vulnerability. Customer should regularly check security updates from NXP and follow up appropriately.

Customer shall select products with security features that best meet rules, regulations, and standards of the intended application and make the ultimate design decisions regarding its products and is solely responsible for compliance with all legal, regulatory, and security related requirements concerning its products, regardless of any information or support that may be provided by NXP.

NXP has a Product Security Incident Response Team (PSIRT) (reachable at PSIRT@nxp.com) that manages the investigation, reporting, and solution release to security vulnerabilities of NXP products.

NXP B.V. — NXP B.V. is not an operating company and it does not distribute or sell products.

Trademarks

Notice: All referenced brands, product names, service names, and trademarks are the property of their respective owners.

NXP — wordmark and logo are trademarks of NXP B.V.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

AMBA, Arm, Arm7, Arm7TDMI, Arm9, Arm11, Artisan, big.LITTLE, Cordio, CoreLink, CoreSight, Cortex, DesignStart, DynamIQ, Jazelle, Keil, Mali, Mbed, Mbed Enabled, NEON, POP, RealView, SecurCore, Socrates, Thumb, TrustZone, ULINK, ULINK2, ULINK-ME, ULINK-PLUS, ULINKpro, µVision, Versatile — are trademarks and/or registered trademarks of Arm Limited (or its subsidiaries or affiliates) in the US and/or elsewhere. The related technology may be protected by any or all of patents, copyrights, designs and trade secrets. All rights reserved.

Bluetooth — the Bluetooth wordmark and logos are registered trademarks owned by Bluetooth SIG, Inc. and any use of such marks by NXP Semiconductors is under license.

i.MX — is a trademark of NXP B.V.

All information provided in this document is subject to legal disclaimers.

© 2023 NXP B.V. All rights reserved.

Contents

1 Introduction 2
1.1 Board overview 2
1.2 i.MX 6ULL EVK contents 3
1.3 i.MX 6ULL EVK board revision history 3
2 Specifications 3
2.1 i.MX 6ULL processor 5
2.2 Boot mode configurations 5
2.3 Power tree 7
2.4 DDR3L DRAM memory 9
2.5 SD card slots (J301, P1701) 10
2.6 eMMC 10
2.7 QSPI Nor flash 10
2.8 NAND flash 10
2.9 Dual Ethernet connector (J1501) 10
2.10 USB PHY connector (J1102, J1101) 11
2.11 Audio input/output connectors (J1401/ P1401/J1402 and J1403) 11
2.12 USB debug connector (J1901) 11
2.13 Bluetooth Connector (J1701) 12
2.14 JTAG connector (J1902) 12
2.15 ARDUINO expansion port (J1703-J1706) 13
2.16 User interface switches 14
2.16.1 ON/OFF button (SW2101) 14
2.16.2 Reset 14
2.16.3 Power switch (SW2001) 14
2.17 Sensors 15
2.17.1 Accelerometer 15
2.17.2 eCOMPASS (U1302) 15
2.17.3 Gyroscope (U1303) 15
2.18 User interface LED indicators 15
2.19 Optional back-up coin cell holder (BT2101) 15
3 PCB information 15
4 EVK design files 16
5 Contents of the Evaluation Kit 16
6 References 16
7 Revision history 17
Legal information 18

Please be aware that important notices concerning this document and the product(s) described herein, have been included in section 'Legal information'.

© 2023 NXP B.V. **All rights reserved.**

For more information, please visit: https://www.nxp.com

Date of release: 2 November 2023
Document identifier: IMX6ULLEVKHUG